



Telecom Customer Churn Prediction System

Machine Learning Capstone Project

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Course: Data Science

Tools: Python • Scikit-Learn • XGBoost • Streamlit

Problem Statement

Customer churn represents one of the most significant challenges facing telecommunications companies worldwide. When customers switch providers, companies face immediate revenue loss and increased acquisition costs—acquiring new customers typically costs five times more than retaining existing ones.

Primary Goal: Develop an accurate machine learning model to identify high-risk customers early, enabling proactive retention strategies and reducing revenue loss.



Key Insights from Data Analysis

Comprehensive analysis of customer data revealed critical patterns influencing churn behavior:



Service Quality Issues

Customers reporting frequent complaints and network problems demonstrate significantly higher churn probability.



Contract Type

Month-to-month contracts exhibit substantially higher churn rates compared to annual or long-term plans.



Billing Patterns

Higher monthly charges correlate directly with increased churn risk, particularly when combined with service issues.



Payment Behavior

Late payments and low engagement levels strongly indicate elevated probability of customer departure.

Critical Finding: Customer behavior, billing patterns, and service quality collectively determine churn probability, providing actionable intervention points.

Machine Learning Models Deployed

Three sophisticated classification models were trained, validated, and rigorously evaluated for churn prediction accuracy:

1

Logistic Regression

Established baseline model for churn classification, providing interpretable coefficients and probability estimates. Serves as foundation for model comparison.

2

Random Forest

Ensemble method leveraging decision trees to capture complex, non-linear patterns in customer behavior and interaction effects.

3

XGBoost

Advanced gradient boosting framework renowned for superior predictive performance, handling imbalanced datasets effectively through iterative optimization.

Selection Criteria: Performance comparison across multiple metrics including accuracy, precision, recall, and F1 score.

Model Performance Comparison

Models evaluated using standard classification metrics to ensure robust predictive capability:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.82	0.79	0.75	0.77
Random Forest	0.86	0.84	0.80	0.82
XGBoost	0.88	0.86	0.83	0.84

Selection Rationale: XGBoost achieved superior overall performance across all metrics, demonstrating strongest predictive power for identifying churn risk.

System Architecture & Workflow

The churn prediction system implements a streamlined four-stage processing pipeline:



User Input

Customer information captured through intuitive Streamlit web interface with validated data entry fields.



Feature Processing

Automated feature engineering transforms raw customer data into predictive features using domain-specific logic.



ML Prediction

Trained models (Logistic Regression, Random Forest, XGBoost) ensemble predicts churn probability with confidence intervals.



Result Visualization

Dashboard displays churn probability, risk classification, and actionable model insights for business interpretation.

Interactive Dashboard Features



Streamlit Application

Developed intuitive web interface enabling seamless churn analysis:

- **Single Customer Prediction:** Manual entry of customer attributes for immediate risk assessment
- **Bulk Prediction:** CSV file upload for batch processing multiple customers simultaneously
- **Model Insights:** Detailed performance metrics and feature importance visualization
- **Risk Segmentation:** Automatic classification into low, medium, or high churn risk categories

Critical Data Patterns Identified



Short Customer Tenure

Recent customers with less than 6 months tenure demonstrate elevated churn probability.



High Complaint Frequency

Multiple service complaints within 30-day period significantly increase departure risk.



Month-to-Month Contracts

Flexible contracts without commitment show 3.5× higher churn than annual agreements.



Payment Delays

Two or more late payments correlate strongly with increased churn likelihood.



Low Engagement

Minimal service usage and account inactivity indicate impending customer departure.

These actionable insights enable targeted retention strategies for high-risk customer segments.

Business Impact & Value Creation



Strategic Benefits

- **Early Risk Detection:** Proactive identification of customers likely to churn before departure
- **Retention Optimization:** Targeted interventions reduce customer loss through personalized offers
- **Revenue Protection:** Even 5% reduction in churn significantly impacts bottom-line profitability
- **Operational Efficiency:** Automated predictions reduce manual analysis time by 80%

ROI Consideration: Customer retention costs substantially less than acquisition, with typical retention ROI of 300%.

Implementation Recommendations

01

Deploy Production System

Integrate churn prediction model into existing CRM and operational workflows for real-time risk monitoring.

02

Target High-Risk Segments

Focus retention resources on customers with frequent complaints, month-to-month contracts, and premium pricing.

03

Personalized Retention Offers

Design targeted interventions including service credits, plan upgrades, dedicated support, or discounted renewals.

04

Continuous Model Monitoring

Track model performance metrics weekly and retrain quarterly with new data to maintain prediction accuracy.

Strategic Imperative: Integrating machine learning-based churn prediction into telecommunications operations delivers measurable reduction in customer attrition and significantly enhances long-term revenue stability.